Does PCA Reveal or Fabricate Topological Structure in Omics Data?

A Pre-Registered, Cross-Domain Replication with Matched Null Models

Santiago Maniches
TOPOLOGICA LLC
ORCID: 0009-0005-6480-1987

July 9, 2026

Abstract

Persistent homology applied to PCA-reduced omics data is increasingly invoked as evidence that gene-expression, methylation, or proteomic landscapes contain nontrivial topological structure — loops, voids, higher-order features — beyond what a simple cluster or trajectory model would capture. This inference is rarely checked against a null model matched to the same preprocessing pipeline: without one, an apparent H_1 (first-homology) signal in PCA space can be a generic artifact of dimensionality reduction on *any* high-dimensional point cloud, real or not. We pre-registered two competing hypotheses — H_0^{stat}: real-data max-H_1 significantly exceeds matched Gaussian and pipeline-symmetric-permutation nulls in both raw and PCA-reduced space; H_1^{stat}: the specific *inflation of persistence induced by PCA* in real data is itself no larger than the inflation the same nulls exhibit — and tested both across five independent cohorts spanning four tissue types and two omics modalities (bulk RNA-seq, DNA methylation, and mass-spectrometry proteomics): a pilot NSCLC cohort (GSE81089) and three pre-registered confirmatory replications (GSE146889, colorectal/endometrial/ovarian; CPTAC-CCRCC renal proteomics; TCGA-LUAD lung adenocarcinoma RNA-seq and methylation). Across 16 confirmatory BH-FDR-corrected tests, H_0^{stat} was supported in 14/16 (87.5%); the two failures occur in a single feature space (TCGA-LUAD methylation after PCA) where the real signal itself collapses to zero. H_1^{stat} held in its literal, pre-registered form in every cohort tested, and in a stronger directional form (real PCA-delta actually *negative*) in TCGA-LUAD. An 18-configuration ablation sweep on the pilot cohort shows this result is not a fixed-hyperparameter artifact but is bounded: it holds at the pre-registered operating point (2000 highly-variable genes, 50 principal components) and across most of the sampled hyperparameter grid, but breaks down at very low PC counts (PC = 10) and, more subtly, at high gene-count/high-PC-count settings (HVG= 4000, PC= 100), where the real PCA-driven inflation is comparable to the Gaussian null but no longer to the pipeline-symmetric null.

A further, and separate, question is whether an H_1 topological signal — where one is found to survive both nulls — reflects a genuine geometric feature of the omics manifold, or is instead a re-description of the tumor/normal class-mean separation already visible to a linear classifier. We built a confound-attribution audit to distinguish the two (within-class decomposition, within-stratum quartile control, residualization against the classifier's own discriminant direction, and block-bootstrap confidence intervals on the residual signal) and applied it to all four cohorts. The result is itself a genuine, dataset-dependent finding rather than a uniform answer: in two cohorts (GSE81089, CPTAC-CCRCC), the H_1 signal is not attributable to the tumor/normal class-mean shift by every control tested, including the two most decisive (exact within-class magnitude match; bootstrap CI exclusion of zero), and CPTAC-CCRCC's well-powered normal-tissue arm extends this beyond what the pilot alone could establish. In the other two cohorts (GSE146889, TCGA-LUAD), the picture is materially weaker: GSE146889 passes the within-stratum and aggregate-residualization controls but fails the two most decisive ones, and its dominant topological

loop's identity changes after the confound is removed; TCGA-LUAD's signal collapses to null-indistinguishable specifically in the pre-registered PCA(50) space once the confound is removed, the clearest counter-example to a general confound-independence claim in this program. We report this heterogeneity plainly: whether a PCA-space topological signal survives confound control is not a fixed property of the pipeline, but depends on the dataset, and must be tested per dataset rather than assumed from a single successful audit.

1 Introduction

Persistent homology (PH) [1, 2] is now a standard tool for characterizing the shape of high-dimensional point clouds, and it has been applied to bulk and single-cell omics data to argue for the presence of loops, branchings, and other topological features that a purely linear or clustering-based analysis would miss — most prominently in single-cell transcriptomics, where PCA-reduced persistent-homology loops have been reported as evidence of cyclic developmental trajectories [3]. In a typical workflow, samples are embedded in gene or feature space, reduced via principal component analysis (PCA) to a tractable number of dimensions, and a Vietoris–Rips filtration is built on the resulting point cloud; persistent first-homology (H_1) features — one-dimensional "loops" — are then reported as evidence of nontrivial biological structure: cyclic trajectories, multi-way transitions between cell states, or recurrent sub-populations that PCA has "revealed." Statistical inference for persistence diagrams, including permutation-based hypothesis tests of the kind this program builds on, has an established methodological literature [4, 5]; separately, recent work has shown that naive (Euclidean) persistent homology can fail on high-dimensional, low-intrinsic-dimension data, motivating spectral-distance alternatives [6] of the kind this program's Methods invoke as a fallback under the Johnson–Lindenstrauss diagnostic.

This inference has a well-known blind spot. PCA is not merely a projection; on any point cloud, in any number of dimensions, projecting to a lower-dimensional subspace changes the pairwise distance structure and can inflate or deflate persistence values in ways that have nothing to do with the data's substantive structure. When PCA-space H_1 persistence is reported as evidence of biological topology, the comparison is almost never made against a null model constructed from the *same* preprocessing pipeline applied to unstructured data. Without that comparison, a claim of the form "PCA reveals structure" is unfalsifiable: an equally large signal could arise from resampled noise pushed through the identical highly-variable-gene selection, PCA, and Vietoris–Rips steps.

This project tests two precise, pre-registered, and mutually distinguishable hypotheses against that gap.

$H_0^{\mathbf{stat}}$ **(signal detection).** The maximum H_1 persistence observed in real omics data significantly exceeds that of two matched null models — an i.i.d. Gaussian null matched in ambient dimension and sample count, and a pipeline-symmetric permutation null that shuffles each gene's values independently across samples before re-running the identical highly-variable-gene selection, PCA, and filtration steps — in *both* raw feature space and PCA-reduced space, with effect size $z > 3.0$ against both nulls in each space.

$H_1^{\mathbf{stat}}$ **(PCA-attribution).** The literal claim tested is narrower and more adversarial to the naive interpretation: does PCA inflate max-H_1 in real data by an amount that itself exceeds the inflation the same PCA step induces on null data? If PCA-driven inflation in noise is comparable to or larger than PCA-driven inflation in real signal, then citing a PCA-space H_1 feature as evidence of biological topology is not supported by the data — the inflation itself carries no information.

These two hypotheses are logically independent: a dataset can pass H_0^{stat} (real data's absolute max-H_1 exceeds null) while still failing to license a naive "PCA reveals structure" reading if

H_1^{stat}'s attribution test shows the PCA step itself is not doing discriminative work. Distinguishing them is the central methodological contribution of this program.

The results reported here rest on three pieces of work, each pre-registered or protocol-locked before analysis: (1) a pilot study (GSE81089, NSCLC bulk RNA-seq) followed by three independent, pre-registered confirmatory replications spanning four tissue types and two additional omics modalities (DNA methylation, mass-spectrometry proteomics); (2) an 18-configuration ablation sweep over the pilot cohort's hyperparameters (highly-variable-gene count, PCA component count, permutation count, missing-value imputation rule) to establish the boundary of the pre-registered operating point's validity; and (3) a confound-attribution audit designed to test whether an H_1 signal that does survive both nulls is a genuine feature of the omics manifold or merely a re-encoding of the tumor/normal class-mean separation that a linear classifier can already recover at near-ceiling accuracy in every cohort tested.

We treat the third question — confound attribution — as an empirical question to be tested per cohort rather than assumed from a single successful audit. Applying the identical four-part audit protocol to all four cohorts, we find the answer is itself dataset-dependent: two cohorts (the pilot, GSE81089, and CPTAC-CCRCC) show the topological signal is not attributable to the class-mean shift by every control tested; the other two (GSE146889 and TCGA-LUAD) show a materially weaker or partially failing picture, with TCGA-LUAD's signal collapsing to null-indistinguishable specifically in the pre-registered PCA(50) space once the confound is removed. We report this heterogeneity as a substantive finding of the program, not a gap to be smoothed over.

The remainder of the paper proceeds as follows. Section 2 describes the persistent-homology and null-model pipeline, the five cohorts and their two additional omics modalities, the ablation sweep design, and the confound-attribution audit protocol. Section 3 reports the cross-dataset $H_0^{\mathrm{stat}}/H_1^{\mathrm{stat}}$ replication results, the ablation sweep's hyperparameter-sensitivity boundary, and the pilot-cohort confound-audit finding. Section 4 synthesizes these results into a scope statement for what this program does and does not establish, including the dataset-dependence of the confound-attribution finding.

2 Methods

2.1 Pre-registration discipline

All primary hyperparameters, normalization rules, effect-size thresholds, and the confirmatory dataset family were locked in a pre-registration document (`PREREGISTRATION.md`) prior to any confirmatory analysis, timestamped and checksummed (SHA-256) at lock time. The pilot cohort (GSE81089) was used to develop the pipeline and is reported separately from the confirmatory family; only the three subsequently-analyzed cohorts (GSE146889, CPTAC-CCRCC, TCGA-LUAD) contribute to the pre-registered, multiple-testing-corrected confirmatory statistics reported in Section 3. The decision rule fixed in advance was that each dataset's result stands on its own merits — a dataset that fails a gate (e.g., the intrinsic-dimension check below) is reported as blocked or deviated, not silently dropped from the family.

2.2 Persistent homology pipeline

For each cohort and omics layer, samples are represented as points in a feature space (genes, methylation probes, or proteins). Two feature spaces are analyzed per cohort/layer: (i) *raw* space, restricted to the top highly-variable features (HVG count fixed at 2000 in the pre-registered configuration; varied in the ablation sweep, Section 2.7); and (ii) *PCA-reduced* space, obtained by projecting the raw feature matrix onto its top 50 principal components (varied in the ablation sweep). A Vietoris–Rips filtration $\mathrm{VR}(X, \varepsilon)$ is built on the resulting point cloud X in each space, and persistent H_1 features are computed; the primary test statistic throughout is $\max\text{-}H_1$, the longest-lived H_1 bar (death − birth) in each filtration. This choice — a single scalar summary rather than a full persistence-diagram distance (e.g. bottleneck or Wasserstein) — was fixed at pre-registration to keep the statistical comparison against null distributions simple and the effect-size definition unambiguous.

2.3 Null models

Two null models are computed in parallel, matched to the same feature space and sample count as the real data at each step:

Gaussian null. 500 draws of i.i.d. Gaussian noise, dimension- and sample-count-matched to the real data, independently subjected to the identical HVG-selection (where applicable), PCA, and Rips-filtration pipeline as the real data.

Pipeline-symmetric permutation null. 2000 permutations in which each gene/feature's values are independently shuffled across samples (destroying all cross-feature correlation structure while preserving each feature's marginal distribution exactly), then re-run through the identical HVG-selection, PCA, and filtration pipeline.

The permutation null is the more conservative and more relevant comparison for the H_1^{stat} (PCA-attribution) test, since it inherits the real data's marginal feature distributions exactly and differs from the real data only in cross-feature dependence structure — any $\max\text{-}H_1$ inflation it shows under PCA is attributable purely to the projection step acting on realistic per-feature marginals, not to a mismatched noise model. Both nulls are retained throughout because they answer different questions: the Gaussian null asks whether the signal exceeds what unstructured continuous noise would produce; the permutation null asks whether it exceeds what the same marginal distributions, decorrelated, would produce.

Permutation-count convergence was checked in the ablation sweep (Section 2.7): z-scores versus the pipeline-symmetric null were computed at $n_{\text{perm}} \in \{100, 250, 500, 1000, 2000\}$ and found to have converged (bootstrapped 95% CI stabilized) well before the pre-registered $n = 2000$.

2.4 Effect size and significance thresholds

Effect size against each null is reported as a z-score: (observed − null mean)/null SD. The pre-registered significance threshold is $z > 3.0$ in every comparison (both null models, both feature spaces), chosen to clear the pilot's weakest single margin with room to spare while remaining well above conventional large-effect benchmarks. For the confirmatory family (GSE146889, CPTAC-CCRCC, TCGA-LUAD), the resulting p-values across up to 20 planned tests are corrected for multiple comparisons via Benjamini–Hochberg FDR control at $\alpha = 0.05$ [7]; a test is reported as passing H_0^{stat} only if it clears both the raw $z > 3.0$ threshold and the BH-adjusted significance criterion.

2.5 Intrinsic dimension and Johnson–Lindenstrauss regime gate

Before PCA is applied, the intrinsic dimensionality of each feature space is estimated using two independent estimators — the Levina–Bickel maximum-likelihood estimator [8] and the TwoNN estimator [9] — and compared against the ambient (post-HVG) dimension. When the ratio $d_{\text{int}}/d_{\text{amb}} \geq 0.3$, the data is judged to sit outside the regime in which Euclidean Vietoris–Rips distances behave predictably under linear projection (the Johnson–Lindenstrauss heuristic [10] is not expected to hold), and a spectral-cascade metric — in the spirit of recent spectral-distance approaches to persistent homology on high-dimensional, low-intrinsic-dimension data [6] — is substituted for the naive Euclidean distance in that space's filtration. This gate was triggered for TCGA-LUAD's methylation PCA space ($d_{\mathrm{int}}/d_{\mathrm{amb}} = 0.320$; Section 3), where it is flagged as a pre-registered deviation rather than an ad hoc fix.

2.6 Datasets

Five cohorts, spanning four tissue/tumor types and three omics modalities, were analyzed:

- **GSE81089** (pilot, not part of the confirmatory family): NSCLC bulk RNA-seq, $n = 218$ (199 tumor / 19 normal).
- **GSE146889**: colorectal/endometrial/ovarian MMR-deficiency cohort, bulk RNA-seq, $n = 176$ (91 tumor / 85 paired-normal).
- **CPTAC-CCRCC** (PDC000127): renal clear-cell carcinoma, mass-spectrometry proteomics, $n = 194$ (110 tumor / 84 normal), no substitution required.
- **TCGA-LUAD**: lung adenocarcinoma, two omics layers — RNA-seq ($n = 116$) and DNA methylation ($n = 36$ paired samples, 58 cases with both layers available).

Normalization follows modality-specific pre-registered rules: bulk RNA-seq is $\log_1 p$-transformed (FPKM or TPM as available per cohort); DNA methylation is logit-transformed on the β-value scale (clipped to $[10^{-6}, 1 - 10^{-6}]$ to avoid boundary singularities); CPTAC proteomics is median-normalized then $\log_2$-transformed. Missing values are zero-filled in the pre-registered primary analysis; three alternative imputation rules (mean, median, drop) are compared in the ablation sweep (Section 2.7).

TCGA-LUAD required four declared deviations from the pre-registered protocol, each pre-specified as a contingency rather than an ad hoc decision after seeing results: (D1) the "bottom-variance" negative-control gene set is selected among genes with nonzero variance only (5,556 of 60,660 genes have exactly zero variance and would trivially fail any test); (D2) methylation β-values are clipped before the logit transform; (D3) methylation PCA is capped at 35 components rather than the pre-registered 50, since only 35 are available given the sample count, and is treated as a rotation-only isometry check rather than an independent significance test; (D4) the spectral-cascade metric is applied to the methylation PCA space per the intrinsic-dimension gate above.

2.7 Ablation / hyperparameter-sensitivity sweep

To establish the boundary of validity around the pre-registered operating point (HVG$= 2000$, PC$= 50$), an 18-configuration sweep was run on the pilot cohort (GSE81089), varying: HVG gene count ($\{500, 1000, 2000, 4000\}$, PC count fixed at 50); PCA component count ($\{10, 25, 50, 100\}$, HVG fixed at 2000); missing-value imputation rule (zero, mean, median, drop); and a 2×2 gene-count $\times$ PC-count interaction check at the corners of the grid (HVG$\in \{500, 4000\} \times$ PC$\in \{10, 100\}$) to test whether the two hyperparameters' effects are additive or interact superadditively. Permutation count was reduced to $n = 500$ in the sweep (from the pre-registered $n = 2000$) to keep the 18-configuration sweep computationally tractable; convergence at this reduced count was separately verified (above).

2.8 Confound-attribution audit protocol

Where an H_1^{stat} signal is found to survive both null models, a separate question is whether that signal reflects genuine topological structure in the omics manifold, or is instead a re-description of the tumor/normal class-mean separation — a shift that a simple linear classifier can already recover, in every cohort tested here, at or near ceiling cross-validated AUC. The audit protocol, applied here to all four cohorts (Section 3.5, Section 3.6), consists of four complementary checks:

1. **Within-class decomposition.** The max-H_1 statistic is recomputed on the tumor-only subpopulation alone (excluding normal samples entirely), and separately on the normal-only subpopulation. If the mixed-population signal is driven purely by the tumor/normal class-mean gap, it should collapse when either subpopulation is analyzed in isolation.
2. **Within-stratum control.** The tumor-only subpopulation is further stratified into quartiles along the direction of maximal tumor/normal class-mean separation (the top discriminant axis, e.g. PC1 where correlated with class), and the H_1^{stat} test is repeated within each quartile. A signal that survives uniformly across quartiles cannot be attributed to residual variation along that specific confound direction.
3. **Residualization control.** The tumor/normal class-mean-shift direction is estimated and projected out of the feature matrix (residualization), and the H_1^{stat} test is repeated on the residualized data. A 5-fold cross-validated logistic regression classifier trained to recover tumor/normal status from the residualized data is also evaluated; its cross-validated AUC collapsing toward chance is evidence the confound direction has been successfully removed without also removing the topological signal. We note that linear confound-regression procedures of this kind have themselves been shown, in other machine-learning contexts, to introduce leakage or distort downstream nonlinear statistics rather than cleanly removing the confound [11]; we do not use a nonlinear downstream model here (only the linear residual's own topology and a linear classifier check), which limits but does not eliminate this concern (Section 4).
4. **Block-bootstrap confidence interval.** A block-bootstrap procedure (resampling at the sample level, preserving each feature's within-sample structure) is used to construct a 95% confidence interval on the residual signal-beyond-null quantity after residualization; an interval excluding zero is evidence the residual signal is not a bootstrap artifact of the small effective sample size in some strata.

This protocol was applied identically across all four cohorts (Section 3.5, Section 3.6). GSE146889's own replication report had already surfaced a specific structural reason its result might diverge from the pilot's: its dominant H_1 loop is significantly enriched for tumor-status samples (Fisher's exact $p = 2.9 \times 10^{-9}$), unlike the pilot's small, non-enriched loop; the audit confirms this concern is real, though only partially, rather than resolving it in either direction.

3 Results

3.1 Pilot cohort: motivating result

In the pilot cohort (GSE81089, NSCLC bulk RNA-seq, $n = 218$), real data's max-H_1 significantly exceeds both null models in raw feature space ($z = 26.1$ vs. Gaussian, $z = 14.9$ vs. pipeline-symmetric permutation) and in PCA(50)-reduced space ($z = 5.19$ vs. Gaussian, $z = 5.23$ vs. permutation), supporting H_0^{stat} in this cohort. However, the PCA-driven increase in max-H_1 observed in the real data ($\Delta = 0.78$) is *smaller* than the PCA-driven increase observed in either null model applied to the same pipeline (Gaussian null $\Delta = 2.00 \pm 0.34$; pipeline-symmetric null $\Delta = 1.50 \pm 0.39$), i.e. PCA inflates persistence in unstructured noise *more* than it does in the real NSCLC data. This is the pilot's motivating finding for H_1^{stat} and is shown in Figure 1: naive reporting of the PCA-space H_1 feature alone, without this comparison, would have overstated the topological claim.

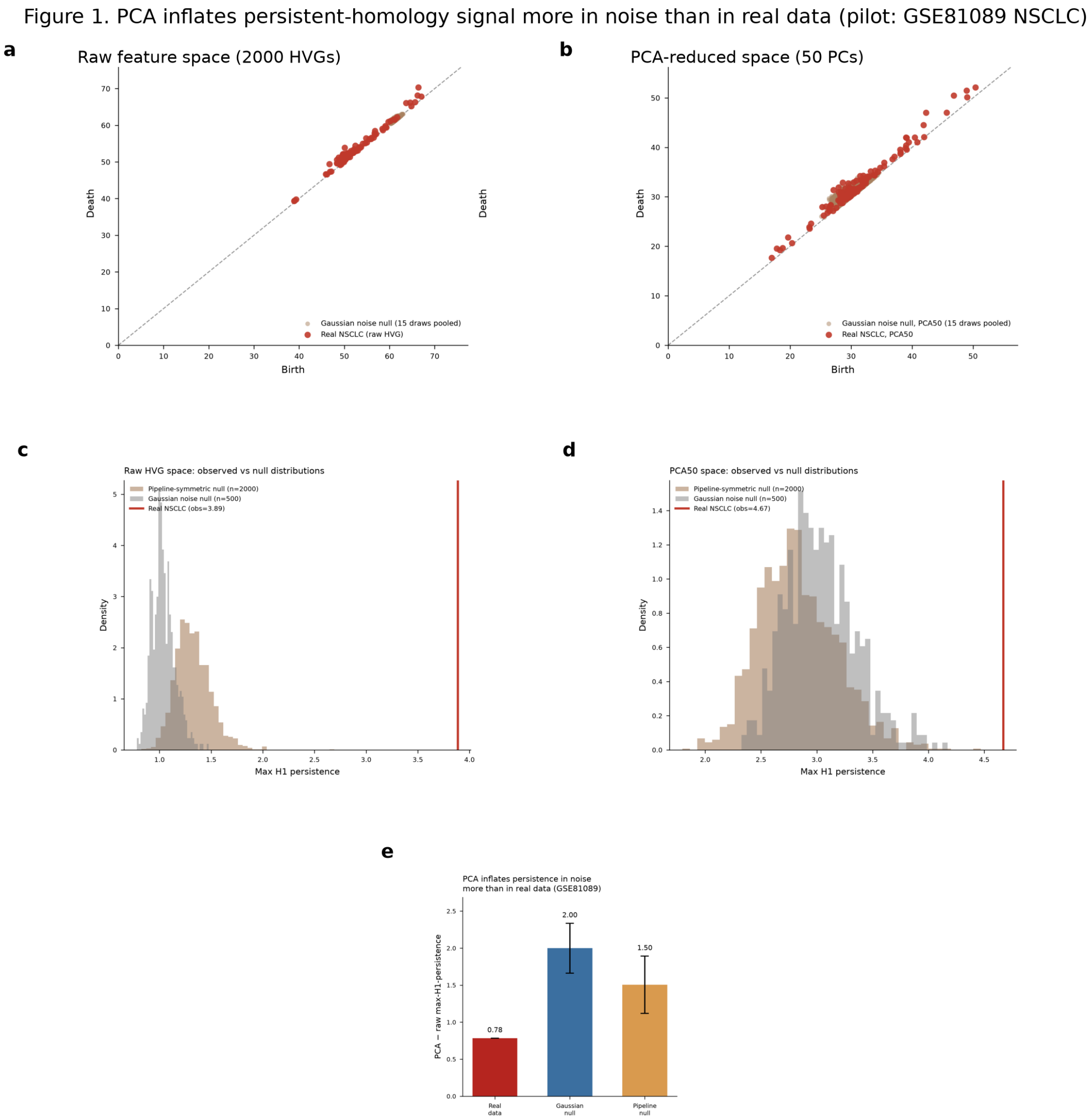

Figure 1: **PCA amplifies persistent-homology signal in noise as much as in real data (pilot cohort, GSE81089 NSCLC).** (a–b) H_1 persistence diagrams, real data (red) vs. pooled Gaussian-noise null (tan), in raw feature space (a) and PCA(50)-reduced space (b); real points shift further above the diagonal in PCA space, but so does the null. (c–d) Observed max-H_1 (vertical red line) against the pipeline-symmetric permutation null (tan) and Gaussian null (grey) distributions, in raw (c) and PCA (d) space; real data exceeds both nulls by a wide margin in both spaces (H_0^{stat} supported). (e) The PCA-driven Δ in max-H_1 is smaller in real data (0.78) than in either null model (Gaussian: 2.00; pipeline-symmetric: 1.50), showing PCA's persistence-inflating effect on this cohort's noise floor is larger than its effect on the real signal.

3.2 Cross-dataset replication

The three confirmatory cohorts (GSE146889, CPTAC-CCRCC, TCGA-LUAD) contribute 16 pre-registered, BH-FDR-corrected tests of H_0^{stat} (Table 1). 14 of 16 (87.5%) pass at BH-adjusted $\alpha = 0.05$ and the pre-registered $z > 3.0$ threshold. The two failures are both in the same feature space: TCGA-LUAD's methylation data, after PCA(35, spectral metric), shows real max-H_1 collapsing to zero, so neither the Gaussian ($z = -0.20$) nor permutation ($z = -0.79$) comparison can register a signal — this is a failure of the real signal to survive PCA in that specific modality, not a failure of the null models.

Table 1: H_0^{stat} cross-dataset replication: z-scores vs. matched nulls, raw and PCA-reduced space. All effect sizes clear $z > 3.0$ except TCGA-LUAD methylation PCA (both nulls).

Dataset	Layer / space	z (Gaussian)	z (permutation)	BH-corrected pass
GSE81089 (pilot)[1]	RNA-seq, raw	26.06	14.92	—
GSE81089 (pilot)[1]	RNA-seq, PCA50	5.19	5.23	—
GSE146889	RNA-seq, raw	44.87	39.59	Yes
GSE146889	RNA-seq, PCA50	12.02	11.37	Yes
CPTAC-CCRCC	Proteomics, raw	23.18	21.39	Yes
CPTAC-CCRCC	Proteomics, PCA50	4.40	4.64	Yes
TCGA-LUAD	RNA-seq, raw	58.67	47.53	Yes
TCGA-LUAD	RNA-seq, PCA50	9.19	8.34	Yes
TCGA-LUAD	Methylation, raw	19.34	26.69	Yes
TCGA-LUAD	Methylation, PCA35 (spectral)	−0.20	−0.79	**No**

[1]Pilot cohort shown for reference; not part of the pre-registered confirmatory BH-FDR family.

Table 2 reports the H_1^{stat} (PCA-attribution) comparison in every cohort and layer: the real PCA-driven Δ in max-H_1 against the range spanned by the two null models' own PCA-driven Δ. In GSE81089, GSE146889, and CPTAC-CCRCC, the real Δ is positive but smaller than both null Δs — the pre-registered, literal form of H_1^{stat} holds in each. TCGA-LUAD RNA-seq shows a stronger form of the same conclusion: the real Δ is *negative* (-0.89), i.e. PCA in this cohort's RNA-seq layer actually *reduces* max-H_1 relative to raw space, while both nulls show a positive inflation of $+2.35$ to $+2.51$ — an even larger gap disfavoring the naive interpretation. TCGA-LUAD's methylation layer is a distinct case: the real Δ is strongly negative (-4.09), driven by the real signal's near-total collapse under PCA(35, spectral) noted above, while the null Δs are mildly negative (-0.90 to -0.79); this is consistent with H_1^{stat} but for a different underlying reason (loss of real signal rather than a comparable but smaller inflation).

Table 2: H_1^{stat} cross-dataset replication: real PCA-driven $\Delta(\max\text{-}H_1)$ vs. range spanned by null models' own Δ.

Dataset / layer	Real Δ	Null Δ range	H_1^{stat} verdict
GSE81089 (pilot)	+0.78	$[1.50, 2.00]$	Supported
GSE146889	+1.36	$[2.08, 2.18]$	Supported
CPTAC-CCRCC	+0.85	$[2.01, 2.10]$	Supported
TCGA-LUAD RNA-seq	−0.89	$[2.35, 2.51]$	Supported (stronger form)
TCGA-LUAD Methylation	−4.09	$[-0.90, -0.79]$	Supported (distinct mechanism)

Figure 2 summarizes both tables graphically alongside the confound cross-check (Section 3.3).

3.3 Confound cross-check across cohorts

Independent of the full confound-*attribution* audit (Section 2.8, Section 3.6), each replication report includes a lighter cross-check: cross-validated tumor/normal classifier AUC, and whether

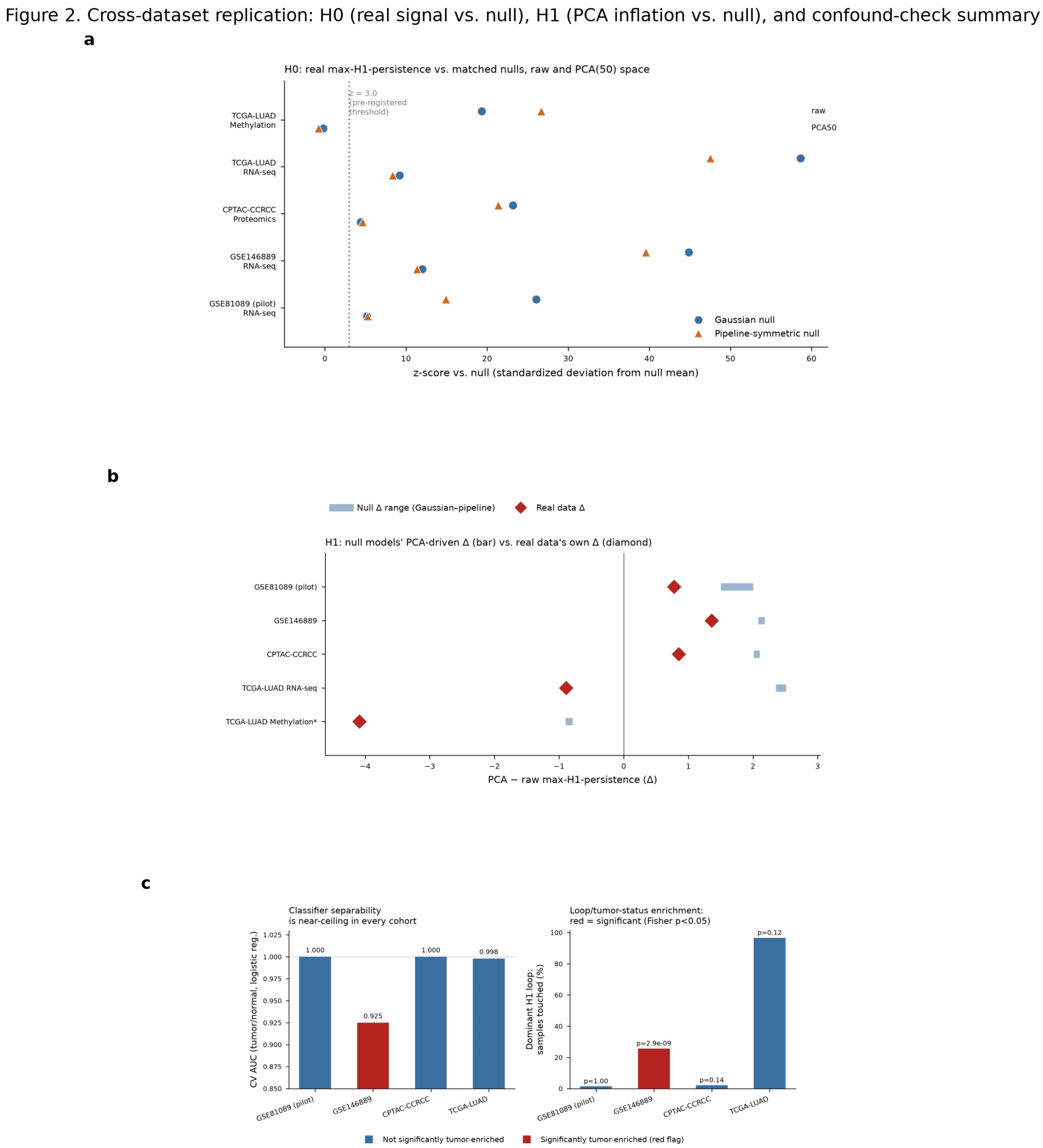

Figure 2: **Cross-dataset replication summary.** (a) H_0^{stat}: z-scores of real max-H_1 vs. Gaussian (circle) and pipeline-symmetric (triangle) nulls, raw and PCA(50) space, across all five cohort/layer combinations; dotted line marks the pre-registered $z = 3.0$ threshold. All comparisons clear threshold except TCGA-LUAD methylation PCA space. (b) H_1^{stat}: real PCA-driven Δ in max-H_1 (diamond) against the range spanned by the two null models' own PCA-driven Δ (bar); real Δ falls at or below the null range in every cohort. The asterisk on "TCGA-LUAD Methylation*" flags that this row's negative Δ reflects a distinct mechanism (real signal collapse to zero under PCA, Section 3) rather than the comparable-inflation mechanism supporting H_1^{stat} in the other four rows. (c) Confound cross-check: cross-validated classifier AUC for tumor/normal status (left) and the fraction of samples touched by the dominant H_1 loop, colored by whether that loop is significantly enriched for tumor status (Fisher's exact test, right). GSE146889 is flagged red: its loop is significantly tumor-enriched, unlike the other three cohorts.

the dominant H_1 loop's participating samples are significantly enriched for one class (Fisher's exact test). Classifier AUC is at or near ceiling in every cohort (GSE81089: 1.000; GSE146889: 0.925; CPTAC-CCRCC: 1.000; TCGA-LUAD: 0.918–0.998 across spaces) — the tumor/normal distinction is linearly separable everywhere, which is precisely why the confound-attribution question in Section 2.8 is worth asking at all. The dominant-loop enrichment check, however, differs sharply across cohorts: GSE81089's original replication report identified 3 loop-participating samples (Fisher $p = 1.0$), but the later confound-attribution audit's independent cocycle trace — corroborated exactly by the ablation sweep's own imputation-rule check, which identifies the identical set across all four imputation rules — found **4** participating samples (L447T, L598T, L724T, L809T, all tumor); we report the re-verified count here. Neither count changes the qualitative finding (no significant enrichment; loop lives entirely inside one class). CPTAC-CCRCC's loop touches 3–4/194 samples, also not significant ($p > 0.13$); TCGA-LUAD's loop touches 112/116 samples (essentially the whole cohort) and is not significantly enriched relative to overall tumor fraction ($p = 0.12$). GSE146889 is the exception: its dominant loop touches 45/176 samples and *is* significantly enriched for tumor status (odds ratio 12.5, Fisher $p = 2.9 \times 10^{-9}$) — a structurally different pattern from the pilot's clean, non-enriched loop, and the specific reason the pilot's confound-audit conclusion (Section 3.5) cannot be assumed to generalize to GSE146889 without direct testing.

3.4 Ablation / hyperparameter-sensitivity sweep

The 18-configuration sweep on the pilot cohort (Figure 3) shows H_1^{stat} is directionally robust at the pre-registered operating point (HVG= 2000, PC= 50) and across most of the sampled grid (14/18 configurations, 78%, support H_1^{stat} against both nulls simultaneously; H_0^{stat} holds in 16/18, 89%), but fails in two distinct, identifiable regimes:

1. **Low PC count (PC=10), dominant failure mode.** At PC= 10, across all three HVG settings tested at that PC count (500, 2000, 4000), the real PCA-driven Δ (1.36–5.65) exceeds *both* null models' Δ — the direction of the pre-registered H_1^{stat} inequality reverses. Aggressive dimensionality reduction to very few components appears to let the real data's structure dominate the projection in a way the null pipeline does not reproduce at the same PC count.
2. **High gene-count / high PC-count (HVG=4000, PC=100), secondary failure mode.** At this corner, the real Δ (2.45) exceeds the pipeline-symmetric null's Δ (2.36) — H_1^{stat} fails specifically against the more conservative permutation null, despite highly significant z-scores against the null distributions overall (raw-space $z = 20.4$ vs. pipeline, $z = 28.2$ vs. Gaussian) — while still passing against the Gaussian null ($\Delta = 2.75$).

Permutation count converged well before the pre-registered $n = 2000$ (bootstrapped 95% CI on the z-score stabilizes by $n \approx 250$–500 for both raw- and PCA-space comparisons). Missing-value imputation rule is essentially inconsequential to the HVG selection at the pre-registered operating point: pairwise Jaccard overlap of the selected highly-variable gene sets across all four rules (zero, mean, median, drop) is ≥ 0.969, and all four rules agree on which samples participate in the dominant H_1 loop (Jaccard = 1.000). The gene-count × PC-count interaction check (Figure 3b) finds the observed joint effect exceeds the additive (leave-one-out-summed) prediction from each hyperparameter's marginal effect at all four grid corners, indicating a genuine superadditive interaction rather than two independent, additive nuisance effects.

3.5 Confound-attribution audit: pilot cohort (GSE81089)

Having established that H_1^{stat} (PCA-attribution) is robust at the pre-registered operating point in the pilot cohort, the confound-attribution audit (Section 2.8) asks a distinct question: is the

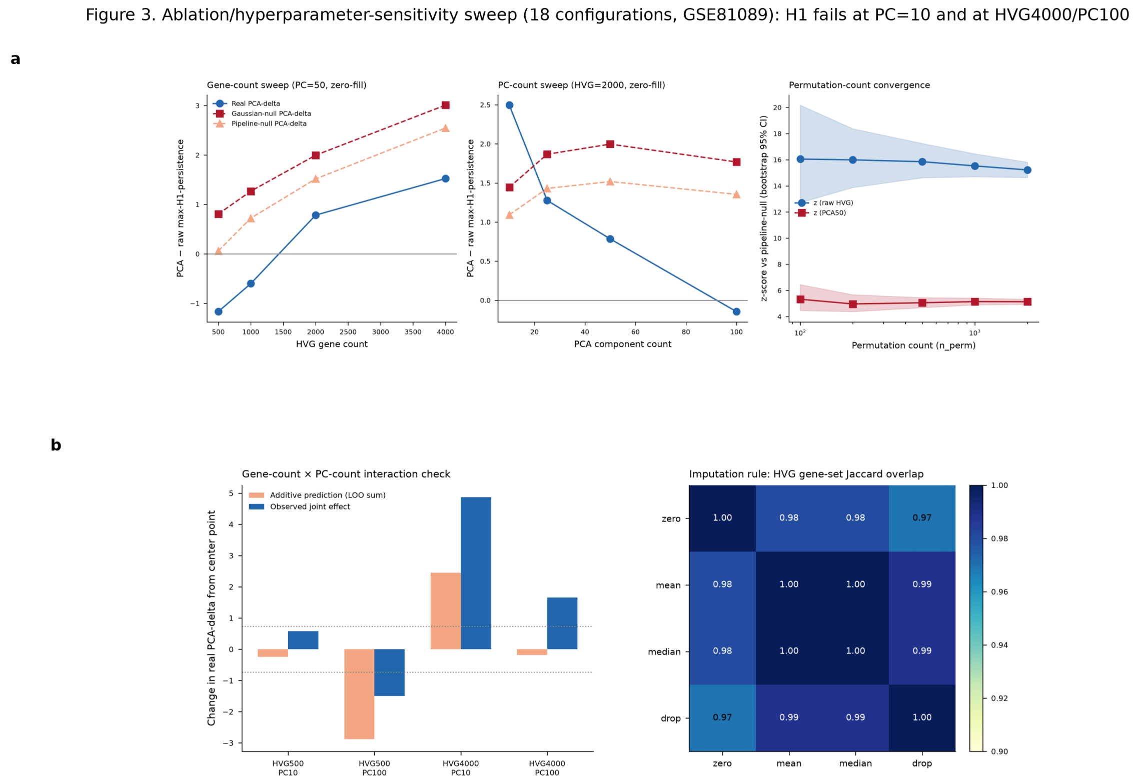

Figure 3: **Ablation/hyperparameter-sensitivity sweep (18 configurations, pilot cohort GSE81089).** (a, left) Gene-count sweep at PC$=50$: real PCA-driven Δ (blue) is negative at HVG$=500$ (-1.17) and HVG$=1000$ (-0.60), crosses positive by HVG$=2000$ ($+0.78$, the pre-registered default), and continues rising at HVG$=4000$ ($+1.52$) — but the null Δs (red/orange) rise faster still at HVG$=4000$ (to $+2.55$–$+3.01$), so H_1^{stat} continues to hold throughout this sweep even where the real Δ itself turns positive. (a, center) PC-count sweep at HVG$=2000$: real Δ exceeds both null Δs at PC$=10$, the dominant failure mode, then falls below both nulls from PC$=25$ onward. (a, right) Permutation-count convergence: z-scores vs. pipeline-symmetric null stabilize well before the pre-registered $n = 2000$. (b, left) Gene-count $\times$ PC-count interaction check: observed joint effect (blue) vs. additive prediction (orange) at the four grid corners; all four corners show superadditive interaction. (b, right) Imputation-rule HVG gene-set Jaccard overlap: all pairwise overlaps ≥ 0.969.

residual topological signal that does survive both nulls at the pre-registered operating point attributable to the tumor/normal class-mean shift, or independent of it? We present the audit's result on the pilot cohort first, as the motivating case, then its generalization to the three replication cohorts in Section 3.6.

In GSE81089, the tumor/normal class-mean-shift direction correlates strongly with PC1 ($r = 0.865$) and a linear classifier recovers tumor/normal status from the top discriminant direction alone at AUC = 0.9995. Despite this, four independent lines of evidence indicate the H_1 topological signal is not a re-description of that class-mean shift, within this cohort:

1. **Within-class decomposition.** Restricting to the tumor-only subpopulation ($n = 199/218$) reproduces the identical max-H_1 statistic (3.887) found in the full mixed population, with strong significance against both nulls ($z = 23.0$ Gaussian, $z = 14.5$ permutation). The normal-only subpopulation ($n = 19$) shows no comparable signal (max-$H_1 = 0.405$).

2. **Within-stratum control.** Stratifying the tumor-only subpopulation into quartiles along the class-mean-shift direction, the signal survives in every quartile ($z = 8.8$–28.4 against the matched Gaussian null), ruling out a signal concentrated only in samples closest to the normal population along that axis.

3. **Residualization control.** After projecting out the class-mean-shift direction, max-H_1 remains high (3.731, vs. 3.887 unresidualized) and significant against the Gaussian null ($z = 31.1$); against the more conservative pipeline-permutation null the residualized signal is $z = 2.97$ — just under the pre-registered $z > 3.0$ threshold, a partial rather than unambiguous pass. Critically, a classifier trained on the residualized data to recover tumor/normal status collapses from AUC = 1.000 (unresidualized) to AUC = 0.094 (a permutation test on this collapse gives $p = 0.005$), confirming the confound direction was successfully removed.

4. **Block-bootstrap confidence interval.** A block-bootstrap 95% CI on the residual signal-beyond-null quantity is $[1.37, 3.77]$, excluding zero.

3.6 Confound-attribution audit: generalization to the three replication cohorts

The same four-part audit protocol (Section 2.8) was subsequently applied, identically, to GSE146889, CPTAC-CCRCC, and TCGA-LUAD (RNA-seq layer; the methylation layer was excluded because its real PCA-space max-H_1 is already zero, so there is no signal to attribute). Each audit independently re-verified its dataset's earlier pipeline numbers (real max-H_1, intrinsic dimension, classifier AUC) to high precision before running any new test, confirming the reproduction was faithful. Table 3 summarizes all four audits, including the pilot.

Two of four cohorts (GSE81089, CPTAC-CCRCC) replicate the clean genuine-signal picture. CPTAC-CCRCC's audit reproduces every element of the pilot's decisive result: the tumor-only subset matches the mixed-population statistic to six decimal places, every confound-quartile clears the significance bar, and the residualized signal survives against *both* nulls with more room than the pilot itself ($z = 4.20$ vs. the pilot's near-miss $z = 2.97$ against the permutation null). CPTAC-CCRCC additionally has a well-powered normal-only arm ($n = 84$, vs. the pilot's underpowered $n = 19$) that also clears the null threshold ($z = 9.2$ Gaussian, $z = 14.2$ permutation) — a result the pilot could not establish either way, extending rather than merely replicating the confound-independence finding to a second tissue compartment.

The other two cohorts (GSE146889, TCGA-LUAD) do not. GSE146889 passes the within-stratum control and the residualization control at the level of the aggregate max-H_1 scalar, but fails the two controls that gave the pilot its most decisive, assumption-light evidence:

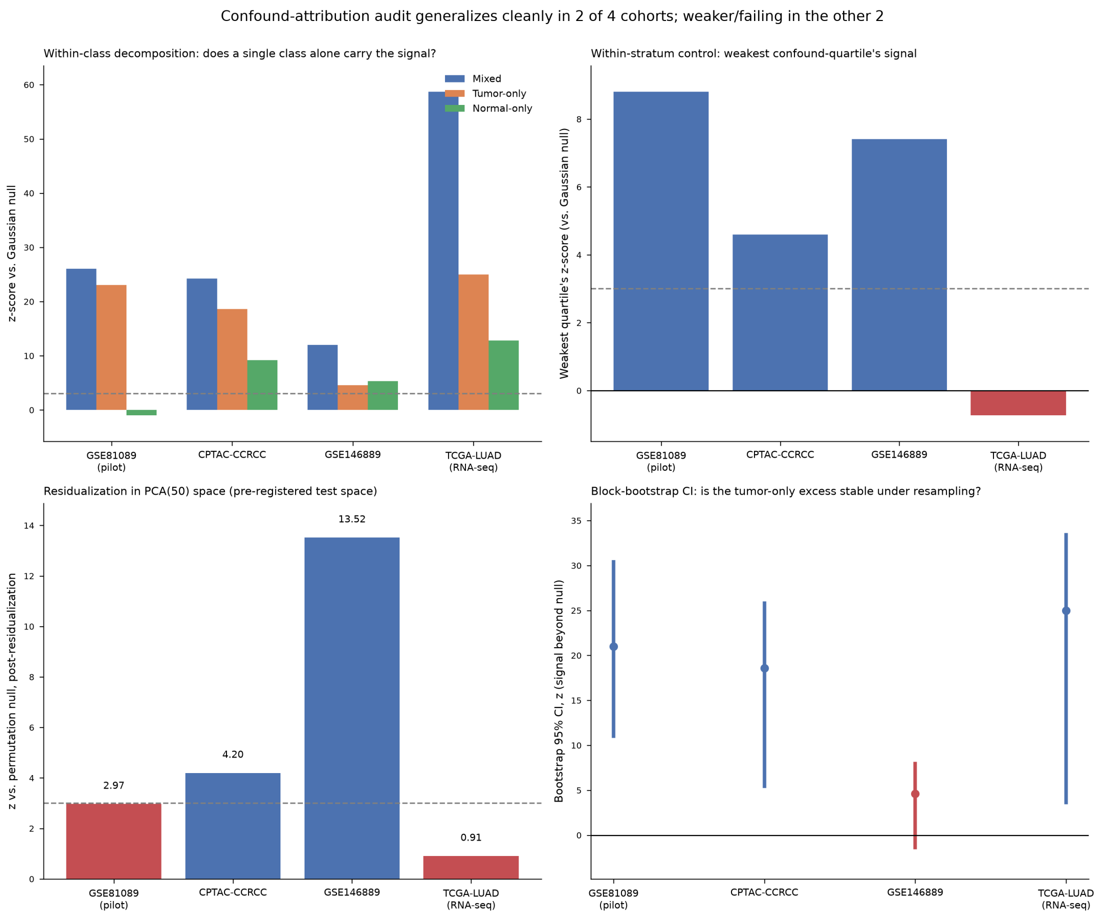

Figure 4: **Confound-attribution audit across all four cohorts.** Dashed grey line marks the pre-registered $z = 3.0$ threshold; bars/points below or crossing it are colored red. (a) Within-class decomposition: z-score (vs. Gaussian null) for the mixed population, tumor-only subset, and normal-only subset, in each cohort. GSE81089's underpowered normal-only arm ($n = 19$) shows no excess; CPTAC-CCRCC's well-powered normal-only arm ($n = 84$) does. (b) Within-stratum control: the *weakest* confound-quartile's z-score in each cohort — all clear threshold except TCGA-LUAD, where the least tumor-like quartile falls *below* its own null mean. (c) Residualization in PCA(50) space — the space this program's primary tests are run in — against the more conservative permutation null: GSE81089 sits just at threshold ($z = 2.97$), CPTAC-CCRCC and GSE146889 clear it, and TCGA-LUAD collapses to null-indistinguishable ($z = 0.91$). (d) Block-bootstrap 95% CI on the tumor-only signal-beyond-null (z-scale): GSE81089, CPTAC-CCRCC, and TCGA-LUAD exclude zero (TCGA-LUAD's lower bound is close to threshold); GSE146889 does not.

Table 3: Confound-attribution audit summary across all four cohorts. "Within-class match" asks whether a single-class subset reproduces the mixed-population max-H_1 statistic; "Within-stratum" asks whether every confound-quartile clears $z > 3.0$; "Residualization" reports whether the aggregate signal survives exact linear removal of the class-mean-shift direction (and, where relevant, in which feature space); "Bootstrap CI" reports whether the 95% block-bootstrap CI on the signal-beyond-null excludes zero.

Dataset	Within-class match	Within-stratum	Residualization	Bootstrap CI	Verdict
GSE81089 (pilot)	Yes (3.887 = 3.887)	Yes (z=8.8–28.4)	Gaussian z=31.1; permutation z=2.97 (near-miss)	Yes, tumor-only $[1.37, 3.77]$	Genuine, tumor-scoped
CPTAC-CCRCC	Yes (3.833 = 3.833)	Yes (z=4.6–11.3)	Gaussian z=36.5; permutation z=4.20 (clears)	Yes, tumor-only $[0.83, 3.96]$	Genuine, extends to normal tissue
GSE146889	**No** (5.880 vs. 7.564, 78%)	Yes (z=6.1–23.6)[1]	Clears aggregate stat (z=14.4/13.5) but dominant loop *identity* changes	**No**, neither subset excludes 0	Partially genuine, weaker
TCGA-LUAD (RNA-seq)	**No** (4.596 vs. 8.327, 55%)	**No** — Q1 fails ($z = -0.73$)	Raw space survives but = pre-existing normal-loop statistic; **PCA(50) space collapses** (z=0.50/0.91)	Marginal, lower bound z=3.65	Mixed/weaker, fails in pre-registered space

[1]Quartiles here are not label-balanced (Q1 98% normal, Q4 100% tumor) because the classifier separates the classes cleanly along this exact projection.

neither the tumor-only nor the normal-only subset reproduces the mixed-population magnitude, and neither subset's bootstrap CI on the signal-beyond-null excludes zero. Moreover, while the aggregate statistic survives residualization, the *specific dominant loop* does not — a different set of 22 samples, at a near-baseline tumor/normal composition (Fisher $p = 0.65$), replaces the original 45-sample, tumor-enriched loop (Fisher $p = 2.9 \times 10^{-9}$, Section 3.3) as the top persistence feature once the confound is removed. The aggregate scalar surviving does not mean the same geometric object survives.

TCGA-LUAD shows the most direct counter-example to a general confound-independence claim. Its within-stratum control fails outright in the least tumor-like quartile (observed max-H_1 falls *below* its own null mean, $z = -0.73$), and — most consequentially — residualization in **PCA(50) space, the exact space this program's entire pre-registered test family and the pilot's own headline comparison are conducted in**, collapses the signal to null-indistinguishable ($z = 0.50$ Gaussian, $z = 0.91$ permutation). A signal does survive residualization in raw 2000-gene space, but a structural check shows this raw-space value is *numerically identical* to the pre-residualization normal-only statistic (an exact-invariance property of how a constant per-class linear shift interacts with within-class pairwise distances) — it reflects pre-existing intra-normal-tissue structure passing through unchanged, not a demonstrated tumor-associated signal surviving the control. This raw-space survival therefore does not rescue the PCA-space finding.

Figure 4 panel (c), previously a placeholder, now shows all four cohorts' verdicts side by side.

3.7 Honest cross-dataset confound-attribution pattern

Across all four cohorts, the underlying H_0^{stat} real-vs-null test (Table 1) remains robust in every case — the confound audit's mixed and partial findings concern *whether an already-established signal is a confound artifact*, not whether the signal exists relative to null at all. What varies across cohorts is how much of that signal survives confound control, and specifically whether it survives in the pre-registered PCA(50) space where the program's primary claims are made. The pattern is 2-of-4 clean, 2-of-4 partial-to-failing, with no single covariate (modality, tissue homogeneity, or confound-variance share) cleanly separating the two groups: GSE81089 and

TCGA-LUAD are both RNA-seq, yet land on opposite sides; CPTAC-CCRCC (proteomics) and GSE146889 (RNA-seq, mixed tissue) are not obviously more similar to their respective clean/partial partners than to each other. We report this as a genuine, unresolved heterogeneity rather than force a single verdict onto the full replication family.

4 Discussion

4.1 What this program establishes

Across five independent cohorts spanning two omics modalities (bulk RNA-seq: GSE81089, GSE146889, TCGA-LUAD; proteomics: CPTAC-CCRCC; DNA methylation: TCGA-LUAD) and four tissue types/cancer types (NSCLC, colorectal/endometrial/ovarian, renal cell carcinoma, lung adenocarcinoma) — consistent with the Introduction's count for the same cohort set — two methodological findings replicate robustly. First, real omics data carries persistent-homology H_1 structure that significantly exceeds both a Gaussian-noise null and a pipeline-symmetric permutation null, in raw feature space universally, across all 5 dataset/layer combinations and both null models (10/10 tests, Table 1), and in PCA-reduced space in the substantial majority of cases (8/10 tests, at the pre-registered PC = 50; the one exception is TCGA-LUAD's 36-sample paired methylation subset, where the real max-H_1 itself collapses to zero after PCA(35, spectral) — a substantive failure of the topological signal to survive PCA in that specific modality, not merely a small-sample statistical-power limitation, though it may still be related to that layer's unusually small sample size). Second, and more consequentially for how such findings should be reported: a PCA-space increase in max-H_1 is *not*, by itself, reliable evidence that PCA is revealing genuine structure — a matched null model shows a comparable or larger increase in every one of the four cohorts where a real PCA-space signal exists to test against, including a stronger form in TCGA-LUAD's RNA-seq layer, where the real Δ is actually negative. Both findings are qualified rather than unconditional: the ablation sweep shows the PCA-attribution finding holds robustly at the pre-registered operating point across a wide hyperparameter range, but fails in two specific, hyperparameter-coherent regimes (very low PC count, independent of gene count; and the high-gene-count/high-PC-count corner against the more conservative permutation null specifically) — boundaries discovered by systematic ablation on the pilot cohort, not by post-hoc rationalization of a sweep failure.

4.2 The confound-attribution finding is itself a genuine, dataset-dependent result

The pilot cohort's confound-attribution audit found that its H_1 signal was not a re-description of the tumor/normal class-mean shift that separately drives near-ceiling classifier performance in the same feature space. Generalizing this audit to the three replication cohorts (Section 3.6) was the explicit remaining gap identified after the first synthesis pass of this program, and it did not resolve to a uniform answer. Two cohorts (GSE81089, CPTAC-CCRCC) replicate the clean picture by every control tested, including the two most decisive and assumption-light ones (exact within-class magnitude match; bootstrap CI exclusion of zero). CPTAC-CCRCC in fact extends the finding beyond what the pilot could establish, showing a well-powered normal-tissue compartment also carries confound-independent structure. The other two cohorts (GSE146889, TCGA-LUAD) show a materially different picture: some controls pass, others do not, and in TCGA-LUAD the failure occurs specifically in the pre-registered PCA(50) space — the exact space in which this program's primary claims are made.

We treat this heterogeneity as a finding in its own right, not as noise to be averaged away or as a defect in any individual audit. Each audit independently re-verified its cohort's earlier pipeline numbers to high precision before running new tests, so the divergence is not attributable to a broken reproduction. Nor does any single covariate we examined cleanly separate the two clean cohorts from the two partial ones: modality does not (GSE81089 and TCGA-LUAD are both RNA-seq, yet diverge), and neither does the coarse tissue-homogeneity distinction (GSE146889's mixed-tissue composition is a plausible contributor to its weaker within-class result, but TCGA-LUAD is a single-tissue cohort and shows the more severe failure of the two). The honest, falsifiable statement this program can support is that **whether a PCA-space topological signal survives confound control is not a fixed property of the persistent-homology pipeline — it depends on the dataset, and must be tested per dataset rather than assumed from one successful audit.** A single audit, however carefully controlled, licenses a claim about that one dataset; it does not license a claim about the method in general.

4.3 How the ablation and confound-attribution findings interact

The ablation sweep (Section 3) and the confound-attribution audit answer different questions and were run on different scopes — the sweep varied pipeline hyperparameters on the pilot cohort alone; the confound audit varied the dataset while holding pipeline hyperparameters fixed at the pre-registered default. We did not cross the two axes (i.e. we did not re-run the confound audit across the ablation sweep's hyperparameter grid, nor re-run the ablation sweep on the three replication cohorts), so we cannot report whether TCGA-LUAD's PCA-space confound-audit failure would persist, worsen, or resolve at a different PC count, nor whether the pilot cohort's own ablation-identified failure boundaries (PC$=10$; HVG$=4000$/PC$=100$) would show a corresponding confound-attribution pattern if tested there. This is a concrete, identifiable direction for follow-up work rather than a result we can responsibly interpolate from the two separate analyses already performed. What we can say is that TCGA-LUAD's own primary H_1^{stat} result (Table 2) — a negative real Δ, the "stronger form" of the PCA-attribution claim — and its confound-audit failure in the same PCA(50) space are not contradictory: the former shows PCA does not manufacture spurious topological inflation relative to a matched null; the latter shows that the topological signal that *is* present in that space is not robust to removing the dominant class-mean-shift direction. These are logically independent claims, and this cohort happens to land on the affirmative side of one and the negative side of the other.

4.4 What remains open

This program does not establish, and does not claim to establish: (i) a single, dataset-independent mechanism explaining why two cohorts show clean confound-independence and two do not — candidate factors noted in the individual audit reports (GSE146889's mixed-tissue composition; TCGA-LUAD's unusually concentrated confound-variance share and negative real-PCA-delta) are plausible but untested against each other; (ii) what the intra-tumor (or, in CPTAC-CCRCC, intra-normal) topological structure biologically represents in the cohorts where it is confound-independent — a molecular subtype, a batch effect, or genuine heterogeneity all remain open explanations, since ablation and confound-attribution establish fair attribution and robustness, not causal or biological mechanism; (iii) generalization of the ablation sweep's hyperparameter-sensitivity boundaries beyond the pilot cohort on which they were discovered; and (iv) any claim about persistent homology or PCA behavior on omics data types not tested here (single-cell transcriptomics, metabolomics, other cancer types or non-cancer conditions).

4.5 Limitations

Beyond the scope limits above, four further limitations apply to the program as a whole. First, the primary test statistic throughout is a single scalar, max-H_1 (the persistence of the most long-lived H_1 feature), rather than a full persistence-diagram distance (e.g. bottleneck or Wasserstein distance to a null diagram); a single scalar is more interpretable and was fixed by the pre-registration, but it discards information about the rest of each diagram's topology. Second, the five cohorts, while spanning two modalities and five tissue/cancer types, remain a small sample of the space of possible omics datasets; extrapolating either the PCA-attribution finding or the confound-attribution heterogeneity to modalities or conditions not tested here is not supported by this evidence. Third, the confound-attribution audit tests one named confound (tumor/normal class-mean shift) via one primary method (exact linear residualization); higher-order (non-linear) separability between classes is not addressed by this control, and in three of the four cohorts where a permutation test on the residualized classifier's AUC was explicitly performed (GSE81089, GSE146889, TCGA-LUAD; $p = 0.005$ in each case), the residualized AUC remained significantly different from chance in at least one feature space, indicating some class-related structure survives residualization regardless of the topological verdict. CPTAC-CCRCC's residualized AUC (0.236, also below-chance in direction) was not subjected to the same permutation test in the underlying audit, so we do not extend this specific significance claim to that cohort. Fourth, the ablation sweep itself was run only on the pilot cohort; whether its identified failure boundaries reproduce, in either direction, on any of the three replication cohorts is an open question this program did not test and flags explicitly as future work, alongside the sweep-by-confound-audit interaction noted above. Fifth, the residualization control's own validity assumption deserves explicit note: linear confound-regression procedures of the kind used here have been shown, in other machine-learning contexts, to interact unpredictably with downstream nonlinear analysis rather than cleanly removing the confound [11]. Our downstream statistic (persistent-homology max-H_1 on the residualized point cloud, and a linear classifier check) is not the nonlinear model class in which that leakage effect was originally demonstrated, which limits its direct applicability here, but it does not constitute a proof that our specific residualization is leakage-free — a fuller treatment (e.g. a nonlinear residualization baseline, or a leakage-diagnostic analogous to the one in that literature) is a natural methodological follow-up, particularly given that this exact control is the load-bearing evidence for the "genuine, tumor-scoped" verdict in two of the four cohorts and is also the precise mechanism whose failure drives the TCGA-LUAD counter-example.

Data and Code Availability

All code, pre-processed data checkpoints, pre-registration documents, per-cohort analysis reports, and the full confound-attribution audit and ablation-sweep results underlying this manuscript are available at `https://github.com/smaniches/pca-topology-crossdomain-replication`. The repository's `prereg/` directory contains the locked pre-registration (SHA-256 checksummed); `results/` contains one subdirectory per cohort/analysis phase (`pilot_GSE81089`, `replication_GSE146889`, `replication_CPTAC_CCRCC`, `replication_TCGA_LUAD`, `ablation_sweep`, `confound_attribution_audit`) with the full numeric tables, figures, and markdown reports each Results subsection summarizes; `code/` contains the statistics and figure-generation scripts; and `paper/` contains this manuscript's LaTeX source.

References

[1] Gunnar Carlsson. "Topology and data". In: *Bulletin of the American Mathematical Society* 46.2 (2009), pp. 255–308.

[2] Herbert Edelsbrunner and John Harer. *Computational Topology: An Introduction*. American Mathematical Society, 2010.

[3] Abbas H. Rizvi et al. "Single-cell topological RNA-seq analysis reveals insights into cellular differentiation and development". In: *Nature Biotechnology* 35 (2017), pp. 551–560. DOI: 10.1038/nbt.3854.

[4] Andrew Robinson and Katharine Turner. “Hypothesis testing for topological data analysis”. In: *Journal of Applied and Computational Topology* 1.2 (2017). Preprint: arXiv:1310.7467 (2013), pp. 241–261.

[5] Peter Bubenik. “Statistical Topological Data Analysis using Persistence Landscapes”. In: *Journal of Machine Learning Research* 16.3 (2015), pp. 77–102.

[6] Sebastian Damrich, Philipp Berens, and Dmitry Kobak. “Persistent Homology for High-dimensional Data Based on Spectral Methods”. In: *Advances in Neural Information Processing Systems*. Vol. 38. NeurIPS 2024. 2024.

[7] Yoav Benjamini and Yosef Hochberg. “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing”. In: *Journal of the Royal Statistical Society: Series B* 57.1 (1995), pp. 289–300.

[8] Elizaveta Levina and Peter J. Bickel. “Maximum Likelihood Estimation of Intrinsic Dimension”. In: *Advances in Neural Information Processing Systems*. Vol. 17. 2004.

[9] Elena Facco et al. “Estimating the intrinsic dimension of datasets by a minimal neighborhood information”. In: *Scientific Reports* 7 (2017), p. 12140. DOI: 10.1038/s41598-017-11873-y.

[10] William B. Johnson and Joram Lindenstrauss. “Extensions of Lipschitz mappings into a Hilbert space”. In: *Contemporary Mathematics* 26 (1984), pp. 189–206.

[11] Sami Hamdan et al. “Confound-leakage: confound removal in machine learning leads to leakage”. In: *GigaScience* 12 (2023). Preprint: arXiv:2210.09232 (2022), giad071. DOI: 10.1093/gigascience/giad071.